

spatiAlert

Getting Started Guide

spatiAlert identifies geographic hotspots of undervaccination using the Getis-Ord G_i^* spatial statistic. This guide walks through data setup, geography and spatial-weights choices, and how to interpret your results — so you can go from a spreadsheet to a hotspot map with confidence.

This app accompanies the analysis published as a research letter in the *New England Journal of Medicine*:

Clusters of Concern — Spatial Link between Childhood Undervaccination and Measles Outbreaks in South Carolina. Published May 20, 2026. *N Engl J Med* 2026;394:2479–2481. DOI: 10.1056/NEJMc2604004

Full text: <https://www.nejm.org/doi/full/10.1056/NEJMc2604004>

Use the “Use paper defaults” switch on the Analysis tab to replicate the published settings exactly.

1. Setting Up

After unzipping the spatiAlert download, you'll have a folder structure like this:

```
spatialert (your chosen folder)
  ■■ run_spatialert - user.R ← open and run this file
  ■■ spatiAlert_Getting_Started.pdf ← this guide
  ■■ spatialert_package/
    ■ spatialert/ (the app itself – don't rename or move this)
```

To launch the app, open **run_spatialert - user.R** in RStudio and set the `dir` variable near the top of the file to the folder that directly **contains** `spatialert_package` — not `spatialert_package` itself, and not the folder inside it.

Example: if you unzipped the download to your Desktop, so that the full path to the package looks like:
/Users/JohnDoe/Desktop/spatialert/spatialert_package/spatialert
then you would set:
`dir <- "/Users/JohnDoe/Desktop/spatialert"`

Then click **Source** (or Ctrl+Shift+S / Cmd+Shift+S). If the path is wrong, the app will tell you specifically what looks off (e.g. pointed too deep, or the folder doesn't exist) rather than a generic error.

2. The Workflow

spatiAlert has three tabs, used in order. All analysis runs locally on your machine — nothing is uploaded to any server.

Tab	What you do
Data & Geography	Upload your data, choose what to analyze, and load or upload a geography (tracts, counties, ZIPs, or a custom shapefile).
Analysis	Choose spatial weights and run the Gi* hotspot statistic (or switch on “Use paper defaults”).
Results & Export	View the map and ranked table, export a CSV, generate a Word report, and copy ready-made methods text.

3. What Data Can spatiAlert Handle?

spatiAlert accepts two data structures. Both are CSV or Excel files.

	Mode A — School / facility level	Mode B — Pre-aggregated
One row per...	school or facility	geographic unit (tract, county, ZIP)
Requires	Latitude & longitude columns for each facility (see below for accepted formats)	A geographic ID column: tract GEOID (11-digit), county FIPS (5-digit), or ZIP/ZCTA

	Mode A — School / facility level	Mode B — Pre-aggregated
What the app does	Spatially joins each facility to the geography you load, then sums or aggregates values within each area	Matches your rows directly to the geography using the ID column — no spatial join needed
Best when...	You have facility addresses/coordinates but no pre-existing geographic assignment	Your source data (e.g. a state health department file) already reports counts or rates by tract/county/ZIP

Mode A details: latitude/longitude

- Two separate numeric columns, e.g. **latitude** and **longitude** (decimal degrees, e.g. 34.0007, -81.0348).
- spatiAlert looks for columns containing “lat” and “lon”/“lng” automatically, but you can remap them.
- Address-only data (no lat/lon) is **not** automatically geocoded in the current version — geocode addresses to lat/lon first (e.g. in Excel, ArcGIS, or a geocoding API) before uploading.
- Rows with missing coordinates are dropped, with a warning showing how many.

Mode B details: geographic ID

Geography	ID format accepted	Example
Census tract	11-digit GEOID (2-digit state + 3-digit county + 6-digit tract)	45079010200
County	5-digit FIPS (2-digit state + 3-digit county)	45079
ZIP / ZCTA	5-digit ZIP Code Tabulation Area	29201

IDs are matched as text, so leading zeros matter — make sure Excel hasn't stripped a leading zero from a FIPS/GEOID column (a common cause of unmatched rows).

4. What Do You Want to Analyze?

Before mapping columns, spatiAlert asks what variable the hotspot analysis should actually run on. This choice is independent of how your data happens to be recorded — you can have percent-formatted data and still choose to analyze the underlying count, or vice versa.

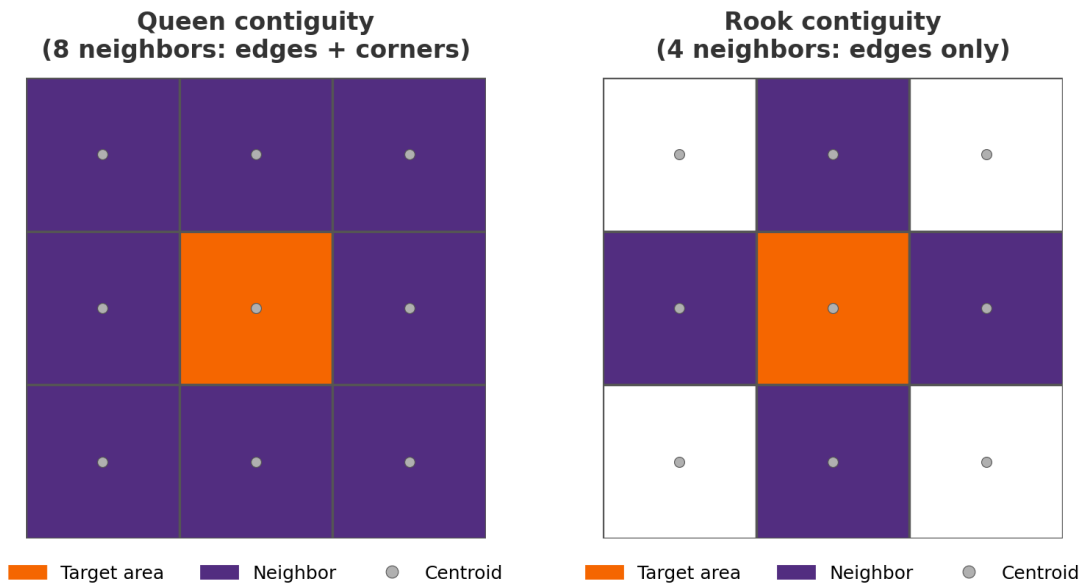
Target	What it highlights	Typically used for
Count of undervaccinated students	Absolute burden — areas with the most undervaccinated students in raw numbers	Resource allocation, outreach targeting (this is what the published NEJM analysis used)
Undervaccination rate	Relative risk — areas with the highest proportion undervaccinated, regardless of population size	Comparing small and large areas fairly; equity analyses
Vaccination rate	The mirror image of undervaccination rate — areas of unusually high coverage	Identifying high-performing areas or benchmarking

Whichever you choose, you'll then tell the app **how that value is recorded** in your file: either as a column you already have directly, or calculated by the app from an eligible-population column plus a vaccinated count or rate. As you map columns, the app shows a live table of exactly what will be calculated using your actual selected column names, plus a “You are analyzing: ____” confirmation banner — so it's unambiguous what the hotspot map represents before and after you run the analysis.

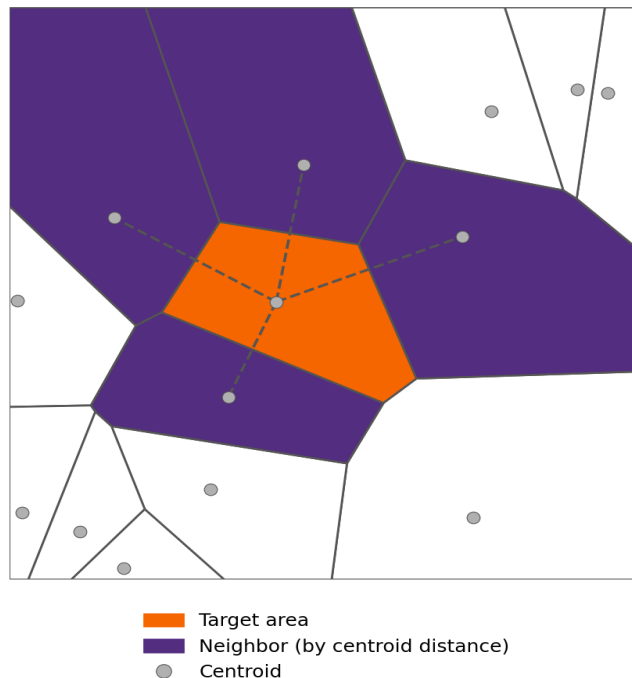
5. Choosing Spatial Weights

Spatial weights define which areas count as each other's “neighbors” — this determines what gets compared to what in the hotspot statistic.

Weights style	How neighbors are defined	When to use it
Queen contiguity	Areas sharing any border point (edge or single vertex)	Simple, standard choice when the geography tiles cleanly with no gaps or islands
Rook contiguity	Areas sharing a border edge (not just a corner)	More conservative version of queen; same caveat about needing full connectivity
K-nearest neighbors (KNN)	Each area's k closest centroids, regardless of shared borders	Guarantees every area has neighbors — essential when geography has islands, non-contiguous areas, or big gaps (e.g. ZIP codes, some tract layouts)



K-nearest neighbors (k = 4)
dashed lines connect the target's 4 closest centroids —
neighbors need not share a border



Note how the KNN diagram uses the same filled-polygon style as queen/rook above, but the purple “neighbor” areas are picked by distance between centroids (dashed lines), not by touching the target area — a KNN neighbor can be non-adjacent, while an adjacent area can be excluded if a closer one exists elsewhere.

Checking whether you can use queen/rook

On the Analysis tab, once your geography loads, spatiAlert automatically shows an FYI panel with the mean, median, and range of neighbor counts under both queen and rook contiguity, and flags any areas with zero neighbors. If either shows 0-neighbor areas, those areas would be silently excluded from a queen/rook

analysis — use KNN instead to keep every area in the analysis.

Why KNN consensus is the default

A single choice of k (e.g. “k=8”) is somewhat arbitrary — results can shift depending on exactly how many neighbors you use. spatiAlert's default mode instead runs Gi* separately across several k values (4, 6, 8, 10, and 12 by default, matching the published analysis) and calls an area a robust hotspot only if it is significant in at least a minimum number of those specifications (2 of 5 by default). This is a robustness check across neighborhood-size assumptions, not a single arbitrary pick.

Binary vs. row-standardized weights

Coding	What it means	Recommended for
Binary (B)	Each neighbor counts equally (weight = 1)	Counts (e.g. number of undervaccinated students) — preserves the additive nature of counts
Row-standardized (W)	Each area's neighbor weights sum to 1 (so more neighbors = smaller individual weight)	Rates or proportions — avoids areas with many neighbors being disproportionately influential

6. The Statistic: Getis-Ord Gi*

spatiAlert uses the **Getis-Ord Gi*** statistic exclusively. For each area, Gi* compares the value there and at its neighbors to the study area as a whole, producing a z-score: positive and statistically significant means a **hotspot** (values are unusually high in that local area); negative and significant means a **coldspot**.

Gi* is well suited to identifying *where burden concentrates* — which is usually the operational question in public health surveillance (“where should we focus outreach?”).

Multiple comparisons correction

Option	What it does	When to use
False discovery rate (FDR)	Controls the expected proportion of false positives among significant results	Recommended default for most exploratory public health analyses
Bonferroni	Very conservative; divides the significance threshold by the number of tests	When you need strict control of false positives, e.g. confirmatory analyses
None	No correction — more areas flagged, some may be false positives	Only with a specific reason; used in the published NEJM analysis, where the emphasis was on exploratory identification of candidate areas rather than confirmatory testing

7. Matching the Published Analysis

Switch on “**Use paper defaults**” at the top of the Analysis tab to lock in:

- **Weights:** K-nearest neighbors, consensus across $k = 4, 6, 8, 10, 12$ (binary coding)
- **Consensus rule:** hotspot/coldspot requires significance in ≥ 2 of the 5 specifications
- **Statistic:** Getis-Ord Gi*
- **Correction:** none
- **Significance threshold:** $\alpha = 0.05$

While the switch is on, these settings are locked so nothing can be changed accidentally — toggle it off to make your own selections again. Note: the paper analyzed the **count** of undervaccinated students. The paper-defaults switch sets the weights/statistic settings above, but the analysis *variable* is set separately on the Data & Geography tab — make sure “What do you want to analyze?” is set to “Count of undervaccinated students” to fully match the published methods.

8. Troubleshooting

The app won't launch / “could not find the app” error

Check the dir path set at the top of run_spatialert - user.R — it should point to the folder that directly **contains** spatialert_package, not the package folder itself. See Section 1 above for a worked example. If the path is close but not quite right, the error message will tell you specifically what looks off.

I selected “percent vaccinated” — what is the hotspot analysis actually run on?

Whatever you selected under “What do you want to analyze?” — not automatically the count. The app always shows a “You are analyzing: ____” confirmation banner on both the Data & Geography and Analysis tabs, plus a live calculation table showing your actual column names, so this is never ambiguous.

My geography has areas with no neighbors

Switch to KNN weights on the Analysis tab, which guarantee every area has neighbors regardless of geographic connectivity. The connectivity FYI panel there tells you whether queen/rook contiguity is even viable for your geography before you pick.

My state only has a few counties. Will Gi* work?

Gi* has limited statistical power with fewer than ~20 units. If your state has very few counties, consider whether tract-level data are available, or interpret county-level results cautiously as a screening exercise.

Column mapping looks wrong / I can't tell what's being calculated

The live “Exactly what will be calculated” panel on the Data & Geography tab shows your actual selected column names plugged into the formula — check that before confirming your data.

Where can I get help beyond this guide?

See the in-app Help tab, or reach out to the app's maintainers directly.

9. References

For readers who want to go deeper on the methodology used in this app:

Clusters of Concern — Spatial Link between Childhood Undervaccination and Measles Outbreaks in South Carolina. Published May 20, 2026. *N Engl J Med* 2026;394:2479–2481. DOI: [10.1056/NEJMc2604004](https://doi.org/10.1056/NEJMc2604004) — *the published analysis this app accompanies.*

Getis, A. and Ord, J.K. (1992). The Analysis of Spatial Association by Use of Distance Statistics. *Geographical Analysis*, 24(3), 189–206. — *the original Gi/Gi* statistic used for hotspot detection in this app.*

Ord, J.K. and Getis, A. (1995). Local Spatial Autocorrelation Statistics: Distributional Issues and an Application. *Geographical Analysis*, 27(4), 286–306. — *distributional properties and inference for Gi*.*

Cliff, A.D. and Ord, J.K. (1981). *Spatial Processes: Models & Applications*. Pion, London. — *foundational reference for spatial weights matrices (contiguity and distance-based, including queen/rook/KNN).*

Benjamini, Y. and Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. — *basis for the FDR multiple-comparisons correction offered in this app.*